

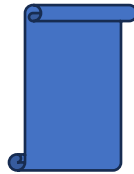
ADIDAS SALES ANALYSIS DASHBOARD



◆ PROJECT OVERVIEW

The **Adidas Sales Analysis** project is an end-to-end business analytics and visualization project focused on understanding sales performance, revenue growth, profitability, product performance, regional distribution, sales methods, and retailer contribution for the period **January 1, 2020 through December 31, 2021**.

The project combines **Python for data exploration, cleaning, exploratory data analysis (EDA), and visualization** with **Microsoft Power BI for interactive dashboard development, DAX calculations, dynamic analysis, and business intelligence reporting**. Five analytical dashboards were developed to provide both high-level KPI monitoring and detailed drill-down analysis across different dimensions of the business. The objective was not only to visualize historical sales data but also to transform the dataset into actionable business insights that can support decisions related to **sales strategy, product portfolio management, regional targeting, retailer performance, inventory planning, and sales-channel optimization**.



◆ THE PROBLEM STATEMENT:

The business requires a comprehensive view of Adidas sales performance to understand **where revenue and profit are being generated, which products and retailers are performing best, how sales vary geographically and which sales methods contribute most to overall performance**.

The analysis addresses the following key business questions:

- How has Adidas revenue and operating profit changed year over year?
- Which products and product categories generate the highest revenue and profit?
- Which states, cities, and regions are the strongest contributors to business performance?
- Which retailers generate the greatest revenue, profit, and unit sales?
- How do **In-Store, Online, and Outlet** sales methods compare?
- How do men's and women's products contribute to overall revenue?
- Which months and years demonstrate significant seasonal sales patterns?
- How does current-year performance compare with the **Previous Year (PY)**?
- Where are the major opportunities for improving revenue, profitability, and sales-channel performance?



◆ DATASET DETAILS

Field	Description
Retailer	Name of the retailer through which the Adidas products were sold.
Retailer ID	Unique identifier assigned to each retailer.
Region	Geographic region associated with the transaction.
State	U.S. state where the transaction occurred.
City	City associated with the transaction.
Gender Type	Identifies whether the product belongs to the men's or women's segment.
Product Category	Category of the Adidas product sold.
Price per Unit	Selling price of one unit of the product.
Units Sold	Number of product units sold in the transaction.
Total Sales	Total sales/revenue generated from the transaction.
Operating Profit	Profit generated after accounting for operating costs.
Operating Margin	Operating profit expressed as a percentage of total sales.
Sales Method	Channel through which the product was sold, such as In-store, Online, or Outlet.
Transaction Date	Date on which the sales transaction occurred.

The analysis is based on the **Adidas Sales Dataset sourced from Kaggle**, containing **9,648 transaction records and 14 fields**. The dataset captures detailed sales information across retailers, products, geographic locations, customer gender, pricing, sales volume, profitability, sales channels and transaction dates. These attributes provide a comprehensive foundation for analyzing **revenue performance, operating profit, product demand, regional trends, retailer contribution and sales-method effectiveness**.

❖ Dataset Structure

- **Rows:** 9,648
- **Fields:** 14
- **Granularity:** Transaction-level sales data
- **Geographic Coverage:** Regions, states and cities across the United States
- **Time Dimension:** Transaction date
- **Business Dimensions:** Retailer, product category, gender type and sales method
- **Key Measures:** Price per unit, units sold, total sales, operating profit and operating margin



◆ DATA EXPLORATION AND CLEANING

The Adidas Sales Dataset was initially explored using **Python** to understand its structure, data types, distributions, missing values, categorical dimensions, numerical measures and potential inconsistencies. **Pandas** was used for data loading, inspection, transformation, and cleaning. The analysis examined fields related to transaction dates, retailers, products, product categories, sales methods, states, regions, cities, units sold, total sales/revenue, and operating profit.

The data preparation process included checking for **missing values, duplicate records, incorrect data types, inconsistent categorical values, and date-related fields**. Transaction dates were converted into appropriate date formats and additional time dimensions such as **Year and Month** were derived to support time-series analysis and year-over-year comparisons.

Exploratory analysis was then performed using **Matplotlib, Seaborn and Plotly** to identify trends, distributions, relationships, outliers, and performance patterns before the data was modeled in Power BI.



◆ WHAT'S INSIDE

1. Adidas Report Overview

The Overview dashboard provides a company-wide summary of performance across **all states, retailers, products, sales methods, and the complete transaction period from January 1, 2020 to December 31, 2021.**

Key KPIs include:

- **Total Revenue:** \$899.90M
- **Previous Year Revenue:** \$182M
- **Revenue Growth:** 394.23%
- **Total Operating Profit:** \$332.13M
- **Previous Year Operating Profit:** \$63.38M
- **Total Quantity Sold:** 2M
- **Total Transactions:** 10K
- **Average Operating Margin:** 42.30%
- **Average Selling Price:** \$363

The dashboard also compares performance across products, retailers, months, product categories, and sales methods.

Men's Street Footwear generated the highest product-level revenue at approximately **\$209M**, while **Women's Athletic Footwear** generated the lowest at approximately **\$107M**. Among retailers, **West Gear** was the leading contributor with approximately **\$242.96M revenue** and **\$85.67M operating profit**, while Walmart recorded the lowest revenue at approximately **\$74.56M**. The monthly revenue trend identified **December 2021 as the highest-performing month**, generating approximately **\$78M**, while June 2020 was the lowest at approximately **\$9M**. At the category level, **Street Footwear** generated the highest revenue at approximately **\$336.83M**, while **Athletic Footwear** generated approximately **\$260.31M**. For sales methods, **In-Store** generated the highest revenue at approximately **\$356.64M**, while Online generated approximately **\$247.67M**.

2. Regional Analysis

The Regional Analysis dashboard focuses on geographic performance across **regions, states and cities**. It enables users to identify geographic sales concentrations and drill down from broader regions into individual states and cities.

New York emerged as a leading state, generating approximately **\$64M in revenue** and **169K units sold**. At the city level, **Charleston** generated approximately **\$39.97M revenue** and **102K**

units sold, making it one of the strongest-performing cities. The **West Region** generated approximately **\$269.94M revenue** and **686,985 units sold**, making it the highest-revenue region based on the reported regional figures. **Seattle** was identified as one of the lower-performing cities, generating approximately **\$26M revenue**.

3. Product Analysis

The Product Analysis dashboard evaluates performance across **individual products, product categories, and gender segments**. **Men's Street Footwear** was the strongest-performing product, generating approximately **\$208.22M in revenue**. At the category level, **Street Footwear** generated approximately **\$336M revenue** and approximately **984K units sold**, making it the leading product category. The analysis also showed that men's products contributed approximately **54% of total revenue (\$485.13M)**, while women's products contributed approximately **46% (\$413.29M)**. The dashboard provides a detailed comparison of **revenue, operating profit, quantity sold, transactions and previous-year performance** to identify high-value products and categories as well as areas requiring further attention.

4. Sales Mode Analysis

The Sales Mode Analysis dashboard evaluates performance across different sales channels, including **In-Store, Online and Outlet**. **In-Store** was the leading sales method by revenue, generating approximately **\$356.64M** and approximately **690K units sold**. Online and Outlet also represented significant contributors to overall sales performance. The dashboard compares sales methods across **revenue, operating profit, units sold, transactions and previous-year performance**, allowing users to understand the contribution and efficiency of each channel. The analysis indicates that **In-Store remained a major contributor to revenue and profitability**, while Online represented an important channel for transaction volume and scalable sales growth.

5. Retailer Analysis

The Retailer Analysis dashboard evaluates the contribution of individual retailers across **revenue, operating profit, quantity sold, transactions, and sales methods**.

West Gear emerged as the strongest retailer, generating approximately **\$243M revenue, 625K units sold**, and approximately **\$85M operating profit**. **Foot Locker** followed with approximately **\$220M revenue** and approximately **\$80M operating profit**. The retailer analysis demonstrates that a relatively small number of major retailers contribute a significant portion of total business performance, while smaller retailers have comparatively lower contributions.

The dashboard would enable stakeholders to compare retailer performance and identify potential opportunities for **retailer strategy, partnership optimization, distribution planning and sales growth**.



◆ DAX FORMULAS & NEW MEASURES

A set of **DAX measures** was created to support dynamic KPI calculations, aggregation, comparison and business analysis within Power BI.

Measures were developed for:

- **Total Revenue**
- **Previous Year Revenue**
- **Revenue Growth %**
- **Total Operating Profit**
- **Previous Year Operating Profit**
- **Operating Profit Growth**
- **Total Quantity Sold**
- **Previous Year Quantity Sold**
- **Total Transactions**
- **Previous Year Transactions**
- **Average Selling Price**
- **Average Operating Margin**
- **Average Revenue by Retailer**
- **Revenue by Product**
- **Revenue by Category**
- **Revenue by Sales Method**
- **Revenue by Region**
- **Revenue by State**
- **Revenue by City**

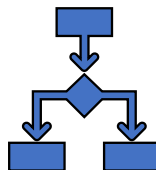
Time intelligence was incorporated to calculate **Previous Year (PY)** performance and support year-over-year comparisons. A **dynamic measure selection parameter** was also created to allow users to switch between key metrics such as **Total Revenue, Total Quantity Purchased, Total Transactions, and Total Operating Profit** within selected visualizations. Additional **Year and Month dimensions** were created to support time-based analysis and allow users to analyze performance at different temporal levels.



◆ VISUALS UTILIZED TO ANALYZE THE DATA

A combination of interactive Power BI visuals was used to communicate the analysis effectively:

- **KPI Cards** — Revenue, operating profit, units, transactions, margins, and growth
- **Gauge Charts** — Current-year revenue/profit compared with previous-year performance
- **Line Charts** — Revenue trends by year and month
- **Clustered Bar Charts** — Product, retailer, category, state, and regional comparisons
- **Combo Charts** — Revenue and profitability comparisons
- **Donut/Pie Charts** — Gender and sales-method contribution
- **Maps / Shape Maps** — Geographic revenue and unit distribution
- **Decomposition Tree** — Region → State → City drill-down analysis
- **Scatter Plots** — Revenue, units, and profitability relationships
- **Matrix Visuals** — Detailed cross-analysis by retailer, product, region, and state
- **Slicers** — Interactive filtering by year, month, region, state, city, retailer, product, and sales method
- **Dynamic Measure Selection** — Switching between key KPIs within selected visuals



◆ STEPS I FOLLOWED TO BUILD THE DASHBOARD

 **Understanding the Data & Business Requirements:** Based on the data, I identified the key areas that needed to be explored, including **revenue, transactions, customers, product categories, payment methods, purchasing modes and time-based trends**. This helped me define the KPIs and visuals required to create a dashboard that would provide both a high-level performance overview and deeper business insights.

 **Building Customer Analysis:** I then focused on understanding **customer purchasing behavior and identifying the best-performing customer**. Rather than evaluating customers

using only revenue, I used multiple measures to provide a broader perspective of **transaction frequency, purchasing quantity and average transaction value**.

 **Creating KPI Cards:** After developing the DAX measures, I created **KPI cards to provide an immediate snapshot of business performance**. The KPI section highlights important metrics such as **Total Revenue, Sales Growth %, Previous-Year Sales, Total Transactions, Total Customers, and Best Customer performance**. These KPIs were positioned prominently so users could quickly understand the overall business performance before exploring the detailed visualizations.

 **Organizing the Dashboard & Adding Analytical Flow :** Once the individual visuals were created, I organized them into a **logical and user-friendly dashboard layout**. The analytical flow was designed to move from **overall KPIs → customer performance → category analysis → payment and transaction modes → time-based trends**. This structure allows users to begin with the overall business picture and gradually explore the factors influencing revenue and transaction performance.



UI DESIGN AND THEME COLORS

The dashboard UI was designed with a **clean, professional, and user-friendly interface** across all five analytical sheets, maintaining a consistent visual experience throughout the project. Two complementary themes were implemented to provide flexibility and visual contrast: a **Light Theme** , featuring a clean white background that enhances readability, clarity, and a modern business-reporting appearance, and a **Dark Theme** , featuring a black background that creates a sleek, contemporary look while improving visual focus on key KPIs, charts, and analytical elements. Both themes were applied consistently across the **Overview, Regional Analysis, Product Analysis, Sales Mode Analysis, and Retailer Analysis** dashboards. The color and layout choices were intentionally kept professional and structured, with clear visual hierarchy, consistent typography, aligned KPI cards, intuitive navigation, and appropriate spacing to prevent visual clutter.   The combination of **light and dark UI themes** allows users to explore the dashboards comfortably while maintaining consistency, readability, and a polished business intelligence experience across the entire Adidas Sales Analysis solution.



◆ KEY INSIGHTS

Revenue & Profitability

- **Total Revenue** reached approximately **\$899.90M**, compared with approximately \$182M in the previous year.
- Revenue increased by approximately **394.23% YoY** based on the dashboard calculations.
- **Operating Profit** reached approximately **\$332.13M**, compared with approximately \$63.38M in the previous year.
- The average operating margin was approximately **42.30%**.

Product Performance

- **Men's Street Footwear** was the strongest individual product with approximately **\$208M** revenue.
- **Street Footwear** was the leading product category with approximately **\$336M** revenue.
- Men's products contributed approximately **54% of total revenue**, compared with 46% from women's products.

Regional Performance

- **West Region** generated approximately **\$269.94M** revenue and approximately **687K** units sold based on the reported regional results.
- **New York** was a leading state, generating approximately **\$64M** revenue.
- **Charleston** was a leading city with approximately **\$40M** revenue.

Sales Method

- **In-Store** generated the highest revenue at approximately **\$356.64M**.
- Online remained an important channel for transaction volume and scalable digital sales.
- Comparing sales methods across revenue, units, profit, and transactions provides insight into channel efficiency.

Retailer Performance

- **West Gear** was the top-performing retailer with approximately **\$243M** revenue and approximately **\$86M** operating profit.
- **Foot Locker** was the second-leading retailer with approximately **\$220M** revenue and approximately **\$80M** operating profit.

Seasonality

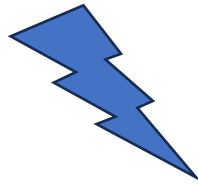
- **December 2021** represented the highest monthly revenue period at approximately **\$78M**.
- **June 2020** represented the lowest reported monthly revenue at approximately **\$9M**.



◆ THE STORYTELLING

The storytelling approach of this project was designed to move from a **high-level business performance overview** to a **detailed root-cause analysis**. The dashboard begins with the overall Adidas business picture, presenting key KPIs such as **Total Revenue, Operating Profit, Units Sold, Transactions, Operating Margin, and Revenue Growth** . From there, the analysis progressively explores **when revenue was generated, where performance was concentrated, which products and categories drove sales, which sales methods were most effective, and which retailers contributed the greatest value**. The story then moves from summary to detail through **Regional, Product, Sales Method, and Retailer Analysis**, allowing users to identify not only *what happened*, but also **where and through which business dimensions it happened**. Interactive **slicers, dynamic measures, geographic maps, decomposition trees, matrices, and comparative visuals** enable stakeholders to move through a structured analytical journey: **Company Performance → Region → State → City → Product → Retailer → Sales Channel** , connecting performance measurement with actionable business insights.

The storytelling also incorporates **Previous Year (PY) comparisons** to provide historical context rather than evaluating current performance in isolation . The substantial increase in **Revenue from approximately \$182M to \$899.90M**, along with the corresponding growth in **Operating Profit from approximately \$63.38M to \$332.13M**, provides the foundation for understanding Adidas's overall growth trajectory. Monthly trends reveal important **seasonal patterns**, while product, regional, retailer, and sales-method analysis identifies the major contributors behind that performance. By connecting these different analytical perspectives, the dashboard transforms a large transactional dataset into a **clear, interactive, and business-focused narrative** that helps stakeholders understand **what happened, why it happened, where the opportunities exist, and which areas require strategic attention**.



🚧 CHALLENGES & HOW THEY WERE ADDRESSED

One of the major challenges was transforming **9,648 transaction-level records across 14 fields** into an analytical structure that could communicate meaningful business insights without overwhelming the user. The dataset contained multiple dimensions—including **retailer, retailer ID, region, state, city, gender type, product category, sales method, and transaction date**—along with financial measures such as **total sales, operating profit, operating margin, price per unit, and units sold**. This complexity was addressed through systematic **data exploration, cleaning, transformation, and dimensional analysis** using Python and Power BI. 🔄 📊

Another challenge was creating reliable **Year-over-Year (YoY) and Previous Year (PY) comparisons** while maintaining consistent KPI calculations across different dashboard views. This was addressed by developing dedicated **DAX measures** for **current-year values, previous-year values, growth percentages, totals, averages, operating profit, transactions, and units sold**. A further challenge was allowing users to analyze multiple business measures without creating separate visuals for every metric. To address this, a **dynamic measure selection parameter** was developed, allowing users to switch between **Total Revenue, Total Quantity, Total Transactions, and Total Operating Profit** within selected visualizations. ⚙️ 📈

The **geographic dimension** also introduced analytical complexity because performance needed to be understood at multiple levels. This was addressed through **interactive maps and decomposition-tree analysis**, allowing users to drill down from **Region → State → City** and identify geographic concentrations and underperforming areas. 🗺️ 🔍 Finally, the dashboard required a balance between **executive-level simplicity and analytical depth**. This was addressed by separating the solution into **five focused analytical views: Overview, Regional Analysis, Product Analysis, Sales Mode Analysis, and Retailer Analysis**, giving each business question an appropriate visual and analytical context. 🎯 📊



◆ WHY THIS PROJECT MATTERS

This project matters because sales data becomes significantly more valuable when it is transformed into **actionable business intelligence** 📊 **rather than simply reported as historical transactions**. The Adidas dataset contains information across revenue, profitability, products, retailers, geography, customer gender segments, sales channels, and time, making it possible to understand the business from multiple perspectives 🔍.

The analysis demonstrates how an organization can use BI to identify **high-value products** 👟, **profitable retailers** 🏪, **geographic opportunities** 🌍, **seasonal demand patterns** 📅, and **channel performance** 🛒. For example, **Men's Street Footwear generated approximately \$208M in revenue**, while **Street Footwear generated approximately \$336M**, identifying major product drivers. **West Gear generated approximately \$243M in revenue and approximately \$86M in operating profit**, highlighting the importance of retailer-level performance. Similarly, **In-Store sales generated approximately \$356.64M**, demonstrating the importance of physical retail channels. These insights can help decision-makers prioritize resources and focus on the areas with the greatest business impact 🎯.

The project also demonstrates the value of combining **Python** 🐍 and **Power BI** 📊 in a modern analytics workflow. Python supports deeper exploration and preparation of the data, while Power BI transforms the results into an interactive environment where business users can explore performance independently. This makes the project relevant not only as a visualization exercise but as an example of how **data analytics can support practical commercial decision-making** 🚀.



🎯 STRATEGIC RECOMMENDATIONS

Based on the analysis, several strategic recommendations can be derived. 🏆 First, Adidas should continue to prioritize **high-performing product categories and individual products**, particularly **Street Footwear and Men's Street Footwear**, while closely monitoring lower-performing categories such as Athletic Footwear. High-performing products can receive greater attention in **inventory planning, promotional campaigns, product placement, and assortment decisions**, while weaker categories should be evaluated for pricing, positioning, demand, and inventory optimization.



Second, retailer performance should be used to guide **distribution and partnership strategies**. With **West Gear and Foot Locker** emerging as major contributors to revenue and operating profit, Adidas could strengthen relationships with high-performing retailers while investigating why lower-performing retailers contribute less. Retailer-level analysis can support differentiated strategies around inventory allocation, promotions, product assortment, and sales targets.



Third, regional and city-level performance should inform **targeted geographic strategies**. Strong-performing locations such as **New York and Charleston** can be prioritized for targeted campaigns and inventory availability, while lower-performing locations such as Seattle can be investigated to determine whether the underlying issue relates to demand, product mix, pricing, retailer presence, or sales-channel availability. The Region → State → City drill-down provides a foundation for making these decisions at a granular level.



Fourth, Adidas should maintain a balanced **omnichannel sales strategy**. Although **In-Store generated the highest reported revenue at approximately \$356.64M**, Online remains important for transaction volume and scalable digital sales. Rather than focusing exclusively on one channel, Adidas can use channel-specific strategies—such as improving digital promotions and customer engagement online while optimizing product availability and in-store experiences.



Finally, the significant increase in revenue and operating profit, combined with visible **monthly seasonality and December 2021's approximately \$78M revenue peak**, suggests an opportunity to strengthen **seasonal planning and forecasting**. Historical monthly trends can be used to anticipate demand, optimize inventory levels, plan promotional campaigns, and allocate resources ahead of peak periods. Future iterations of the dashboard could extend this analysis into **predictive forecasting and demand planning**, enabling Adidas to move from descriptive reporting toward more proactive decision-making.